

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	176091	49.092947
1	[Git, Change, Create]	51952	14.483856
2	[Git, Change, Delete]	27797	7.749610
3	[Java, ByteCode, Member, Method]	13494	3.762033
4	[Java, ByteCode, Parameter]	13293	3.705996
5	[Git, Commit]	13133	3.661389
6	[File, Git]	11021	3.072578
7	[Git, Change, Rename]	10654	2.970261
8	[Java, ByteCode, Bound, ParameterizedType]	7279	2.029335
9	[Java, ByteCode, Bound]	7240	2.018462
10	[Java, ByteCode, Member, Field]	3609	1.006164
11	[Java, ByteCode, Bound, WildcardType]	2947	0.821603
12	[Java, Value, ByteCode, Annotation]	2931	0.817142
13	[Xml, Element]	2162	0.602751
14	[Java, ByteCode, Member, Method, Constructor]	2120	0.591041
15	[Xml, Text]	1450	0.404250
16	[Java, ByteCode, Bound, TypeVariable]	1111	0.309739
17	[Java, ByteCode, Member, Method, Lambda]	968	0.269872
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	888	0.247568
19	[Type, File, Java, ByteCode, Class]	846	0.235859
20	[Json, Key]	702	0.195713
21	[Value, Json, Scalar]	685	0.190973
22	[Java, Value, ByteCode, Primitive]	672	0.187349
23	[Type, File, Java, ByteCode, JavaType]	649	0.180937
24	[Java, ByteCode, Member, Method, GenericDeclar...	578	0.161142
25	[Type, File, Java, ByteCode, ExternalType]	411	0.114584
26	[Git, Change, Copy]	334	0.093117
27	[Author, Git, Person]	299	0.083359
28	[Value, Array]	287	0.080014
29	[Committer, Git, Person]	252	0.070256

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

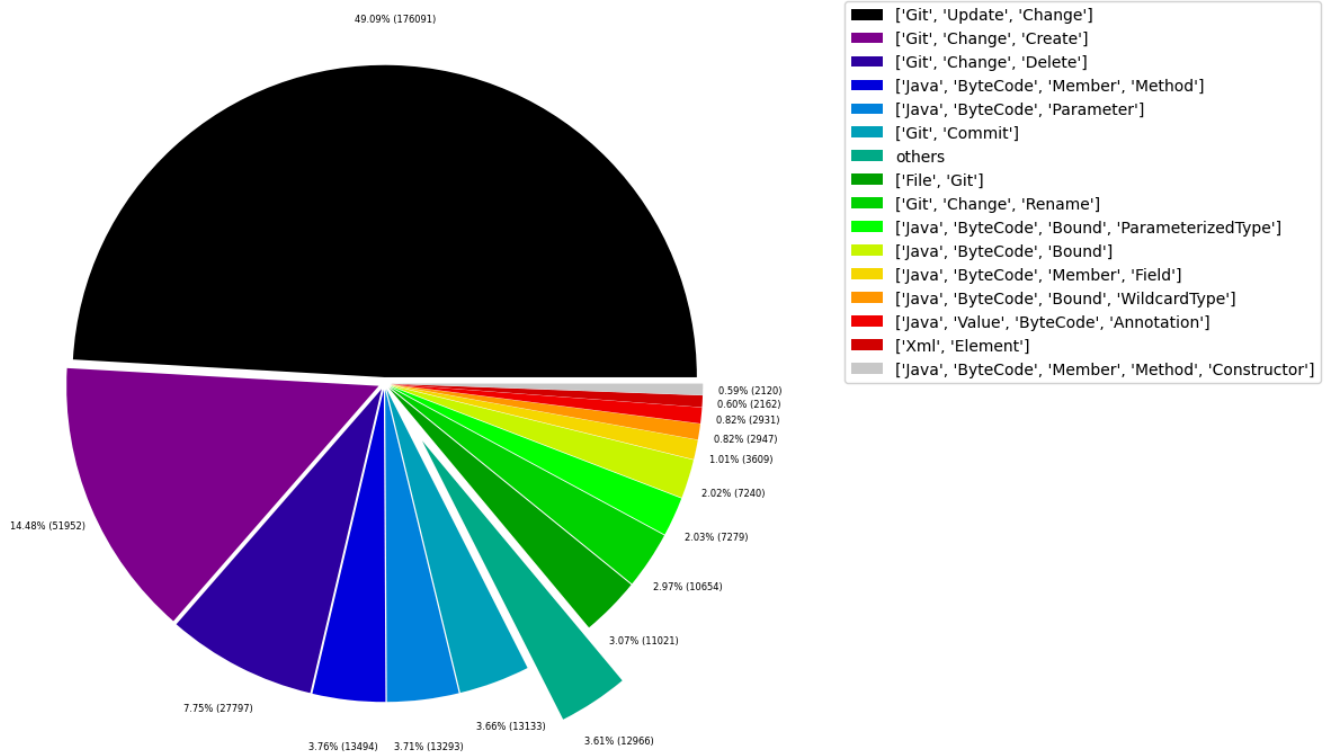


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000279
1	[Package, File, Json, NPM]	1	0.000279
2	[Repository, File, Git]	1	0.000279
3	[File, TS, Scan]	1	0.000279
4	[File, Json]	2	0.000558
5	[File]	3	0.000836
6	[Java, ByteCode, Member, Method, GenericDeclar...	4	0.001115
7	[Maven, Exclusion]	5	0.001394
8	[Value, Array, Json]	6	0.001673
9	[Dependency, NPM]	7	0.001952
10	[Type, File, Java, ByteCode, Void]	9	0.002509
11	[File, Maven, Xml, Pom, Document]	9	0.002509
12	[Java, ManifestSection]	9	0.002509
13	[File, Java, Manifest]	9	0.002509
14	[Artifact, File, Jar, Archive, Zip, Java]	9	0.002509
15	[File, Java, ServiceLoader]	10	0.002788
16	[File, Java, Properties]	12	0.003346
17	[Maven, ExecutionGoal]	16	0.004461
18	[Maven, PluginExecution]	16	0.004461
19	[Xml, Attribute]	18	0.005018
20	[jQAssistant, Rule, Concept]	19	0.005297
21	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.005855
22	[Maven, Configuration]	21	0.005855
23	[Maven, Plugin]	21	0.005855
24	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.006133
25	[Type, File, Java, ByteCode, Enum]	28	0.007806
26	[Type, File, Java, ByteCode, PrimitiveType]	30	0.008364
27	[Xml, Namespace]	36	0.010037
28	[Type, File, Java, ByteCode, Annotation]	44	0.012267
29	[Git, Branch]	46	0.012824

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (less than 0.5% overall)

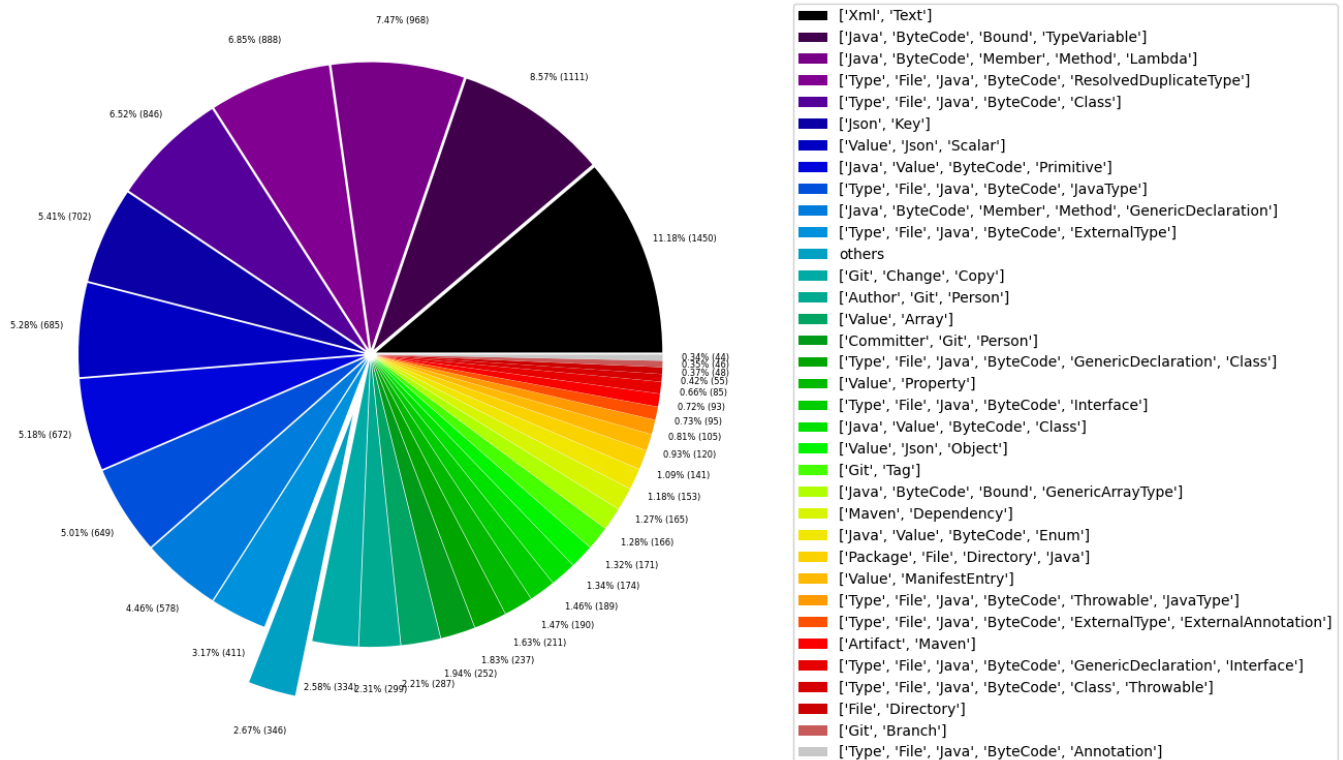


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

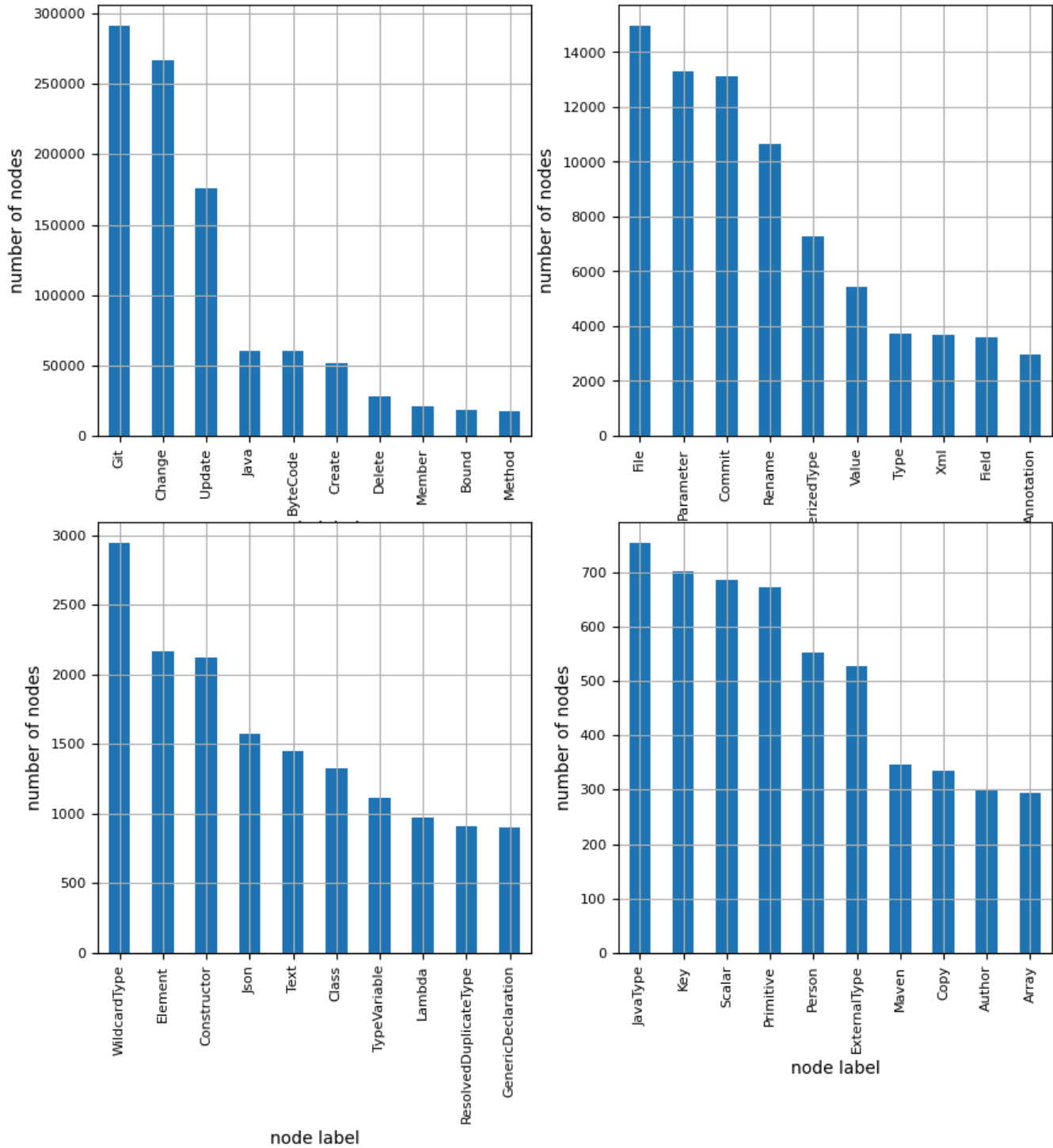
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	291751	81.338151
1	Change	266828	74.389792
2	Update	176091	49.092947
3	Java	60659	16.911308
4	ByteCode	60469	16.858337
5	Create	51952	14.483856
6	Delete	27797	7.749610
7	Member	20773	5.791368
8	Bound	18743	5.225418
9	Method	17164	4.785204
10	File	14982	4.176877
11	Parameter	13293	3.705996
12	Commit	13133	3.661389
13	Rename	10654	2.970261
14	ParameterizedType	7279	2.029335
15	Value	5428	1.513289
16	Type	3715	1.035716
17	Xml	3675	1.024564
18	Field	3609	1.006164
19	Annotation	2975	0.829409
20	WildcardType	2947	0.821603
21	Element	2162	0.602751
22	Constructor	2124	0.592156
23	Json	1570	0.437705
24	Text	1450	0.404250
25	Class	1327	0.369958
26	TypeVariable	1111	0.309739
27	Lambda	968	0.269872
28	ResolvedDuplicateType	910	0.253702
29	GenericDeclaration	904	0.252029
30	JavaType	754	0.210210
31	Key	702	0.195713
32	Scalar	685	0.190973
33	Primitive	672	0.187349
34	Person	551	0.153615
35	ExternalType	527	0.146924
36	Maven	346	0.096462
37	Copy	334	0.093117
38	Author	299	0.083359
39	Array	293	0.081686

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

<Figure size 640x480 with 0 Axes>

Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1113016

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	266828	23.973420
1	MODIFIES	266828	23.973420
2	UPDATES	176091	15.821066
3	CREATES	62940	5.654905
4	DELETES	38451	3.454667
5	INVOKES	36776	3.304175
6	COMMITTED	26266	2.359894
7	DEPENDS_ON	22492	2.020816
8	OF_TYPE	21920	1.969424
9	DECLARES	21234	1.907789
10	OF_RAW_TYPE	17353	1.559097
11	HAS_PARENT	15887	1.427383
12	HAS	14429	1.296387
13	HAS_COMMIT	13133	1.179947
14	RETURNS	12844	1.153982
15	HAS_FILE	11021	0.990192
16	RENAMES	10654	0.957219
17	READS	9378	0.842575
18	HAS_ACTUAL_TYPE_ARGUMENT	8407	0.755335
19	HAS_NEW_NAME	6269	0.563244
20	OF_GENERIC_TYPE	5996	0.538716
21	RESOLVES_TO	5270	0.473488
22	SIMILAR	4078	0.366392
23	WRITES	3944	0.354352
24	CONTAINS	3936	0.353634
25	RETURNS_GENERIC	3593	0.322817
26	ANNOTATED_BY	2919	0.262260
27	REQUIRES	2230	0.200357
28	HAS_FIRST_CHILD	2162	0.194247
29	HAS_LAST_CHILD	2162	0.194247

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

<Figure size 640x480 with 0 Axes>

Relationship types (more than 0.5% overall)

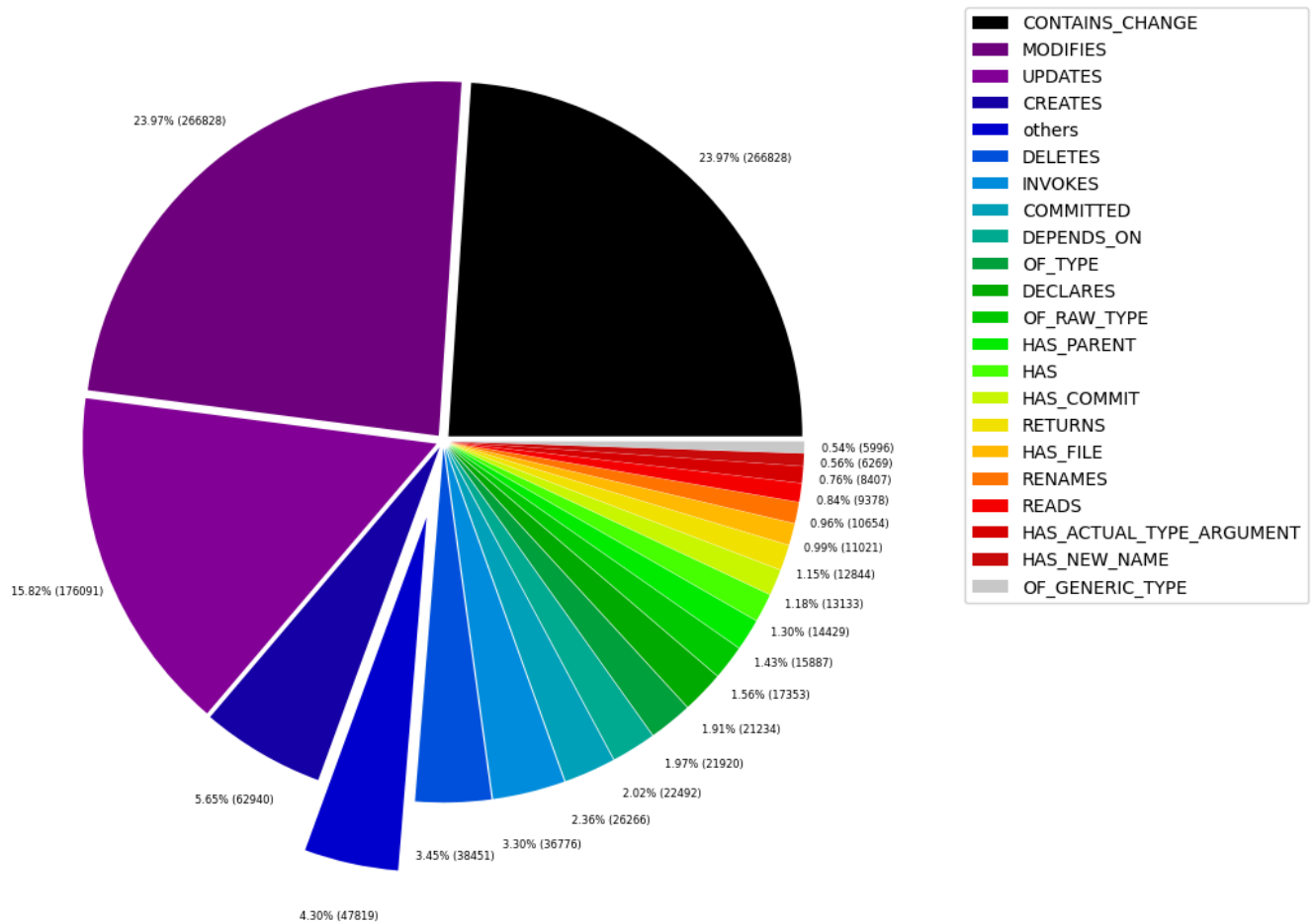


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

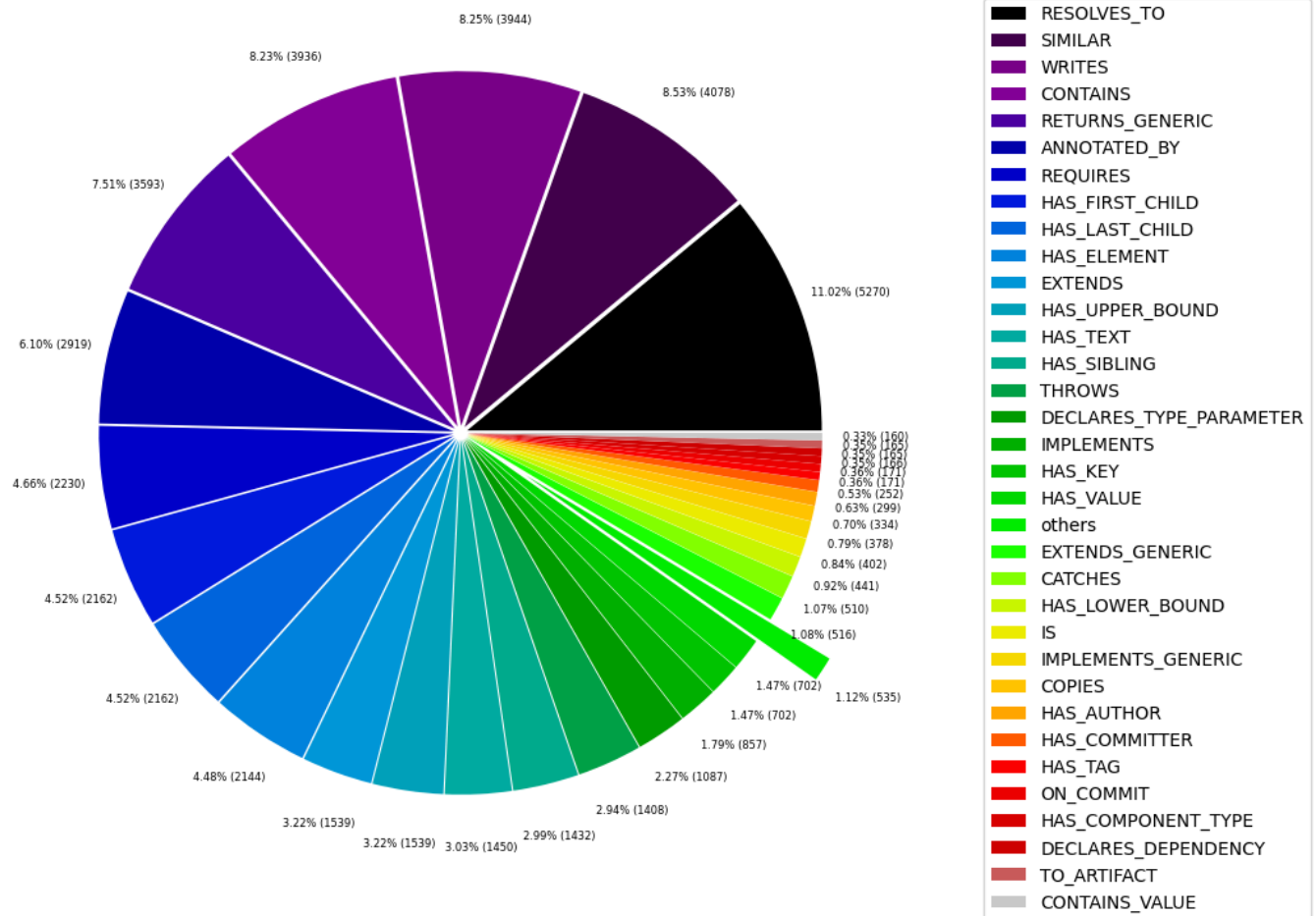
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000090
1	THROWS_GENERIC	5	0.000449
2	EXCLUDES	5	0.000449
3	DECLARES_DEV_DEPENDENCY	7	0.000629
4	DESCRIBES	9	0.000809
5	HAS_ROOT_ELEMENT	11	0.000988
6	HAS_GOAL	16	0.001438
7	HAS_EXECUTION	16	0.001438
8	OF_NAMESPACE	18	0.001617
9	HAS_ATTRIBUTE	18	0.001617
10	INCLUDES_CONCEPT	19	0.001707
11	IS_ARTIFACT	21	0.001887
12	HAS_CONFIGURATION	21	0.001887
13	USES_PLUGIN	21	0.001887
14	REQUIRES_TYPE_PARAMETER	24	0.002156
15	REQUIRES_CONCEPT	28	0.002516
16	DECLARES_NAMESPACE	36	0.003234
17	HAS_DEFAULT	39	0.003504
18	HAS_BRANCH	46	0.004133
19	HAS_HEAD	47	0.004223
20	COPY_OF	127	0.011410
21	CONTAINS_VALUE	160	0.014375
22	DECLARES_DEPENDENCY	165	0.014825
23	TO_ARTIFACT	165	0.014825
24	HAS_COMPONENT_TYPE	166	0.014914
25	ON_COMMIT	171	0.015364
26	HAS_TAG	171	0.015364
27	HAS_COMMITTER	252	0.022641
28	HAS_AUTHOR	299	0.026864
29	COPIES	334	0.030009

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	176091		13133
1	[Git, Update, Change]	MODIFIES	[File, Git]	176091		176091
2	[Git, Update, Change]	UPDATES	[File, Git]	176091		176091
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	51952		13133
4	[Git, Change, Create]	CREATES	[File, Git]	51952		51952
5	[Git, Change, Create]	MODIFIES	[File, Git]	51952		51952
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	27797		13133
7	[Git, Change, Delete]	DELETES	[File, Git]	27797		27797
8	[Git, Change, Delete]	MODIFIES	[File, Git]	27797		27797
9	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	22590		13404
10	[Git, Commit]	HAS_PARENT	[Git, Commit]	15878		13133
11	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	13133		1
12	[Committer, Git, Person]	COMMITTED	[Git, Commit]	13133		252
13	[Author, Git, Person]	COMMITTED	[Git, Commit]	13133		299
14	[Repository, File, Git]	HAS_FILE	[File, Git]	11021		1
15	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	10654		13133
16	[Git, Change, Rename]	CREATES	[File, Git]	10654		10654
17	[Git, Change, Rename]	MODIFIES	[File, Git]	10654		10654
18	[Git, Change, Rename]	DELETES	[File, Git]	10654		10654
19	[Git, Change, Rename]	RENAMES	[File, Git]	10654		10654
20	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	8517		13404
21	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	8446		13404
22	[File, Git]	HAS_NEW_NAME	[File, Git]	6269		11021
23	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	6156		13293
24	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3465		7240
25	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3110		7279
26	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2947		7279
27	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, Bound, ParameterizedType]	2695		13293
28	[Java, ByteCode, Member, Method, Constructor]	WRITES	[Java, ByteCode, Member, Field]	2519		2120
29	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, TypeVariable]	2474		7279

Graph Density

```
total_number_of_nodes (vertices): 358689
total_number_of_relationships (edges): 1113016
-> total directed graph density: 8.6510037796739e-06
-> total directed graph density in percent: 0.00086510037796739
```